

The multiplicity theorem without ordering hypotheses:

$$\int_{(0,b]^{D+2}} u^h f(\sqrt{n} u^k) du \sim C n^{-p/2} (\log n)^m$$

Claude, with Daniel Murfet

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Abstract

For arbitrary exponent vectors $k, h : \text{Fin}(D + 2) \rightarrow \mathbb{N}$ with $k_i > 0$, let p be the minimum of the candidate exponents $q_i = (h_i + 1)/k_i$ and $m + 1$ the number of i attaining it. Then the box integral at scale $\sqrt{n}$ is asymptotically $C n^{-p/2} (\log n)^m$ for some $C > 0$ (`boxIntegralFin_isEquivalent_general`). The coordinates are sorted with `Tuple.sort`, the first index attaining the minimum is a `Finset.min'`, the multiplicity is computed through `Finset.card_equiv` and `Fin.card_Ici`, and the two cases (all exponents equal; some strictly larger) are units 50 and 58. This closes the leading-term theory of `thm:TaylorTree` for constant ξ and monomial η in the chart geometry. Zero `sorry/axiom`.

1 Setting

Unit 58 needed a relabelling putting the exponents in the sorted shape. `Tuple.sort` applied to $-q$ gives a permutation σ with $q \circ \sigma$ antitone; then $p = q(\sigma(\text{last}))$ is the minimum, the set of indices j with $q(\sigma j) = p$ is an upper set, hence the interval $[i_0, \text{last}]$ for its least element i_0 , and everything below i_0 is strictly above p .

2 The things formalised

```
noncomputable def finExp {d : } (k h : Fin d → ) (i : Fin d) : := ((h i : ) + 1) / k i
noncomputable def sortPerm {d : } (k h : Fin d → ) : Equiv.Perm (Fin d) :=
  Tuple.sort fun i => -(finExp k h i)
theorem sortPerm_antitone {d : } (k h : Fin d → ) : Antitone (finExp k h sortPerm k h)
theorem boxIntegralFin_isEquivalent_general ( a b : ) (h : 0 < ) (hb : 0 < b) (D : )
  (k h : Fin (D + 2) → ) (hk : i, 0 < k i) :
  (p : ) (m : ) (C : ), 0 < C ( i, p finExp k h i)
  m + 1 = (Finset.univ.filter fun i => finExp k h i = p).card
  (fun n => boxIntegralFin a b (Real.sqrt n) k h) ~[atTop]
  fun n => C * (n ^ (-(p / 2))) * Real.log n ^ m
```

3 Proof strategy

With i_0 the least index of the minimal set S : $S = \text{Ici } i_0$ (antitone plus minimality), so the multiplicity is $D + 2 - i_0$. If $i_0 = 0$ all exponents equal p and unit 50 applies through the bridge of units 54 and 58, with $m = D + 1$ and constant $(\prod k_i^{-1}) A_{p-1} / (2^{D+1} (D + 1)!)$. Otherwise $i_0 = r + 1$, $m = D + 1 - i_0$, and the sorted theorem in its $r + 2 + m = d$ form

(`boxIntegralFin_isEquivalent_of_perm'`) applies with the sortedness, block and strictness hypotheses read off through `chartExp_toNatFun_eq_finExp`; the constant is positive because each factor is (`genScale_pos`, `genTower_logBase`).

4 Roadblocks and resolutions

- `Tuple.sort` lives in `Mathlib.Data.Fin.Tuple.Sort`, not in the import closure.
- `linarith` does not unfold the definition `sortPerm`; `simp` only `[sortPerm]` first.
- Membership of the last index in the minimal set is `rfl` after unfolding the filter; `simp` stopped at the `set-bound` variable.
- The unit 58 theorem fixed the dimension as `r + 2 + m` in the types; a variant with the equation `r + 2 + m = d` (rewritten on the $\mathbb{N}$ side only) avoids transporting along a propositional equality of types.

5 What was Mathlib, what was new

Mathlib: `Tuple.sort`, `Tuple.monotone_sort`, `Finset.min'` and its API, `Finset.card_equiv`, `Fin.card_Ici`, `Fin.le_last`. New: the hypothesis-free multiplicity theorem.

6 Lessons

Keep type-level dimensions fixed in the workhorse theorems and add a thin variant taking the dimension as a propositional equation; sorting arguments then live entirely on the $\mathbb{N}$ side.

7 Outcome

Units 44–59 formalise the leading-term content of `thm:TaylorTree` in the chart geometry for constant ξ and monomial η : for every exponent vector the box integral is $C n^{-p/2} (\log n)^m$ with p half the minimal candidate exponent and $m + 1$ its multiplicity, and in the sorted form the constant and a complete polynomial-in- $\log n$ expansion are explicit.